

OmniShield 360

AI-Powered Healthcare Claims Protection Platform

UiPath AgentHack 2026 Submission

Executive Summary

OmniShield 360 is an enterprise healthcare claims governance platform built using:

- UiPath Maestro
- UiPath Agents (Python SDK)
- UiPath Robots
- UiPath Automation Cloud
- Human-in-the-Loop (HITL)
- AI Decisioning

The platform protects healthcare payers from fraud, compliance violations, duplicate claims, deceased-member claims, coordination-of-benefits errors, and prior authorization failures.

Instead of treating claims as isolated transactions, OmniShield 360 treats every claim as a governed case that moves through intelligent AI checkpoints orchestrated by Maestro.

Hackathon Goal

Primary Goal

Win the UiPath AgentHack by demonstrating how Agents, Robots, and Maestro work together to solve a real-world enterprise problem with measurable business impact.

Secondary Goals

Demonstrate:

- Agentic AI
- Human-in-the-loop governance
- Enterprise compliance
- Multi-agent orchestration
- End-to-end automation
- Healthcare industry applicability

Problem Statement

Healthcare organizations lose billions annually due to:

Problem	Annual Impact
Fraudulent Claims	\$100B+
Duplicate Billing	\$14B+
Administrative Waste	\$250B+
Prior Authorization Delays	\$1.3B+
Deceased Member Processing	Multi-million dollar legal exposure

Most current systems:

- React after payment
- Operate in silos
- Lack governance
- Depend heavily on manual review

Organizations need proactive prevention.

Solution Overview

OmniShield 360 introduces a four-stage protection framework.

Stage 1

Compliance Shield Agent

Purpose:

Prevent claims involving deceased members.

Checks:

- SSDI
- DMF
- Internal eligibility systems

Actions:

- Block processing
- Trigger Maestro case escalation
- Create audit trail
- Notify compliance teams

Expected Result:

Prevent regulatory violations before payment occurs.

Stage 2

Deduplication Intelligence Agent

Purpose:

Detect duplicate claims and billing fraud.

Technique:

- SHA256 fingerprinting
- Member matching
- Provider matching
- Service-date matching

Actions:

- Auto-reject duplicates
- Route suspicious claims
- Generate fraud alerts

Expected Result:

Reduction in duplicate payouts.

Stage 3

Referral Validation Agent

Purpose:

Verify prior authorization and PCP referral requirements.

Checks:

- Referral existence
- Effective dates
- Provider network participation

Actions:

- Approve
- Reject
- Escalate

Expected Result:

Reduced authorization leakage.

Stage 4

Coordination of Benefits Agent

Purpose:

Determine primary payer responsibility.

Logic:

- Birthday Rule
- Dual Coverage Analysis
- Member Eligibility Review

Actions:

- Assign payer order
- Prevent incorrect payments
- Generate audit evidence

Expected Result:

Reduced overpayments.

Architecture

Patient Claim

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UiPath Maestro Case

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Compliance Agent

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Dedup Agent

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Referral Agent

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COB Agent

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Human Review (if required)

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Robot Execution

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Payment Decision

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Audit Archive

UiPath Components

Maestro

Acts as:

- Central orchestration layer

- Case management platform
- Decision tracking system
- Escalation engine

Responsibilities:

- State transitions
 - Audit history
 - SLA tracking
 - Human approvals
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Agents

Compliance Agent

Detects:

- Deceased members
- Compliance risks

Deduplication Agent

Detects:

- Duplicate claims
- Fraud patterns

Referral Agent

Validates:

- PCP referrals
- Prior authorizations

COB Agent

Determines:

- Primary payer
 - Secondary payer
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Robots

Perform:

- EHR lookups
 - Legacy system access
 - Data extraction
 - Form updates
 - Notification delivery
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Human In The Loop

Certain cases require manual review.

Examples:

- Deceased match confidence below threshold
- Suspected fraud
- Missing referral documentation
- Coverage conflicts

Human reviewers can:

- Approve
- Reject
- Request additional information

All actions remain fully auditable.

Security

HIPAA-focused controls:

- PII masking
- Data tokenization
- Audit logging
- Access controls
- Secure orchestration

Protected Data:

- SSN
- Member ID

- Date of Birth
 - Claim Information
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Testing Strategy

Five end-to-end scenarios.

Scenario 1

Clean Claim

Expected:

Approved

Scenario 2

Duplicate Claim

Expected:

Fraud Alert

Scenario 3

Deceased Member

Expected:

Compliance Block

Scenario 4

Missing Referral

Expected:

Authorization Failure

Scenario 5

Dual Coverage

Expected:

COB Assignment

Expected Business Impact

Financial Impact

For a 500,000-member health plan:

Duplicate Claims

Estimated Savings:

\$2M-\$5M annually

Prior Authorization Automation

Estimated Savings:

\$3M-\$8M annually

Compliance Risk Reduction

Avoidance:

\$6M+ legal events

Coordination of Benefits

Estimated Savings:

\$2M-\$7M annually

Total Estimated Impact

\$15M-\$28M annual value

for a mid-sized payer.

Why This Project Wins

This solution demonstrates all major UiPath innovation pillars:

- ✓ Agents
 - ✓ Maestro
 - ✓ Robots
 - ✓ Human-In-The-Loop
 - ✓ Enterprise Governance
 - ✓ Real Business Value
 - ✓ Production Architecture
 - ✓ Healthcare Impact
 - ✓ Compliance
 - ✓ Measurable ROI
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Demo Flow

1. Create claim
2. Launch Maestro case
3. Compliance agent evaluates
4. Dedup agent evaluates
5. Referral agent evaluates
6. COB agent evaluates
7. Human review triggered if required
8. Robot completes downstream actions
9. Final decision generated
10. Audit evidence stored

Total Demo Time:

4-5 minutes

Winning Message

Healthcare organizations lose billions every year because critical decisions are fragmented across systems and teams.

OmniShield 360 combines UiPath Maestro, Agents, and Robots into a unified governance platform that prevents fraud, ensures compliance, reduces administrative waste, and delivers measurable financial impact before money leaves the organization.

The result is a governed, explainable, AI-powered healthcare operations platform capable of saving a mid-sized payer \$15M-\$28M annually while improving member outcomes and regulatory compliance.

Built for AgentHack. Built for Enterprise. Built for Scale.